DC Power Supply Controller

Project Documentation & User Reference

Arduino Uno | 4-Channel Relay Module | PWM Output Control

1. Project Overview

This project is an Arduino Uno-based DC power supply controller that manages a load through four relays and a PWM output. It automates a timed ramp-up, hold, and ramp-down cycle with safety features including an emergency shutdown switch, button lockout, and a test mode that disables one relay.

The system is designed to be reliable and repeatable — every cycle follows the same sequence, and all critical safety interlocks are hardware-enforced through dedicated switches.

1.1 Cycle Summary

Each run follows this sequence:

- Relays turn on one at a time (250 ms apart, relay 1 through 4)
- PWM output ramps from 0% to 100% over 5 seconds
- Output holds at 100% for either 30 seconds or 2 minutes (selectable via switch)
- PWM output ramps back down from 100% to 0% over 5 seconds
- Relays turn off one at a time in reverse order (relay 4 through 1)
- 5-second button lockout prevents immediate restart

2. Hardware & Parts List

Component	Qty	Notes
Arduino Uno (Rev 3)	1	Microcontroller — must support PWM on pin 9
Lystin 8-Channel Relay Module	1	5 V coil, active-LOW trigger, optocoupler isolated. Channels 1–4 in use; channels 5–8 unused/reserved.
HiLetgo BTS7960 Motor Driver	1	43 A peak H-bridge. Receives PWM from pin 9. Powers the DC load. Requires separate 6–27 V supply.
Momentary Push Button	1	Start button — normally open, wired to GND
Toggle / Rocker Switch ×3	3	Shutdown (pin 3), Test Mode (pin 8), Hold Time (pin 11) — SPST
DC Power Supply	1	6–27 V DC; current rating must match load. Connected to BTS7960 motor power terminals.
Connecting Wire / Jumpers	—	22 AWG solid or 24 AWG stranded recommended
Breadboard or PCB	1	For prototyping or permanent installation

Note: All switches use the Arduino's internal pull-up resistors. No external pull-up resistors are required — simply connect one terminal to the Arduino pin and the other to GND.

3. Hardware Details

3.1 Lystin 8-Channel Relay Module

The Lystin module contains eight independently controlled relays, each with optocoupler isolation between the Arduino control signal and the relay coil. Only channels 1–4 (IN1–IN4) are used in this project; channels 5–8 are wired but left inactive and may be reserved for future expansion.

Specification	Value
Coil voltage	5 V DC (powered from Arduino 5 V rail)
Trigger logic	Active LOW — LOW = relay energised, HIGH = relay off
Contact rating	AC 250 V / 10 A or DC 30 V / 10 A
Driver current (each)	15–20 mA per channel
Isolation	Optocoupler (PCF817) on each channel
Channels in use	1, 2, 3, 4 (IN1–IN4 → Arduino pins 4–7)
Channels unused	5, 6, 7, 8 (reserved for future use)
Board dimensions	134 × 53 × 17 mm

> **Note:** The JD-VCC jumper on the relay module links the relay coil power to the logic VCC. Leave the jumper in place for a single shared 5 V supply. Remove it and provide a separate 5 V supply to JD-VCC if full optical isolation from the Arduino is required.

3.2 HiLetgo BTS7960 43A Motor Driver

The BTS7960 is a fully integrated high-current H-bridge motor driver based on two BTS7960H half-bridge ICs. Both RPWM and LPWM control pins are connected, but this project uses one-direction mode only: RPWM receives the PWM signal from Arduino pin 9, and LPWM is held permanently LOW (tied to GND). This keeps the output in a single fixed direction at all times — output power is controlled entirely by the 0–100% duty cycle on RPWM. Built-in protection covers overcurrent, overtemperature, overvoltage, and undervoltage conditions.

Specification	Value
Motor supply voltage	6–27 V DC (separate supply, not Arduino 5 V)
Logic supply (VCC)	5 V DC (from Arduino 5 V pin)
Peak current	43 A
PWM frequency	Up to 25 kHz (Arduino pin 9 runs ~490 Hz by default)
Control input level	3.3–5 V logic compatible
RPWM (pin 9)	PWM signal from Arduino — controls output power (0–100%)

Specification	Value
LPWM	Held LOW (tied to GND) — fixes output to one direction only
R_EN / L_EN	Enable pins — both must be HIGH (connect to Arduino 5 V) to enable output
Protection	Overcurrent, overtemperature, overvoltage, undervoltage, short circuit
Board dimensions	50 × 50 × 43 mm

Note: The motor power supply (VM+) must be connected directly to the BTS7960 power terminals — never to the Arduino. The Arduino supplies only the 5 V logic line, the RPWM PWM signal, and the R_EN/L_EN enable signals. If R_EN or L_EN are left floating, the driver output will be disabled.

4. Pin Assignments

Pin	Signal	Type	Description
2	switchPin	Digital Input	Main start button (active LOW, internal pull-up)
3	shutdownPin	Digital Input	Emergency shutdown switch (active LOW, internal pull-up)
4	relayPin[0]	Digital Output	Relay 1 — active LOW (LOW = relay energised)
5	relayPin[1]	Digital Output	Relay 2 — active LOW (disabled in test mode)
6	relayPin[2]	Digital Output	Relay 3 — active LOW
7	relayPin[3]	Digital Output	Relay 4 — active LOW
8	testModePin	Digital Input	Test mode switch — LOW = test mode ON (relay 2 skipped)
9	outputPin	PWM Output	PWM signal to load device (0–255 duty cycle)
11	holdTimePin	Digital Input	Hold time selector — LOW = 30 s, HIGH = 2 min

Note: Relay module inputs are active-LOW. The Arduino drives them LOW to energise a relay and HIGH (default) to de-energise it. All four relay pins are set HIGH at startup so all relays start in the OFF state.

5. Switch Wiring & Behaviour

All four switches follow the same wiring convention: one terminal to the designated Arduino pin, the other terminal to GND. The Arduino pin is configured as INPUT_PULLUP, so it reads HIGH when the switch is open and LOW when closed.

Switch	Pin	Behaviour
Start Button	2	Momentary press (LOW pulse) starts a cycle when idle. During HOLD, resets the hold timer. During RAMP DOWN, reverses ramp direction back up.
Shutdown Switch	3	Toggle LOW to trigger immediate shutdown: PWM cut to 0, all relays off instantly. 15-second lockout follows. After lockout, hold LOW for 5 seconds to reactivate.
Test Mode Switch	8	Toggle ON (LOW) while idle to start a cycle with relay 2 skipped and the hold phase running indefinitely. Toggle OFF to begin graceful ramp-down.
Hold Time Selector	11	Read once at cycle start. LOW = 30-second hold. HIGH (open) = 2-minute hold. Changing mid-cycle has no effect until the next cycle.

CAUTION: The shutdown switch is not debounce-protected against false triggers at power-on. Ensure the switch is in the open (HIGH) position before powering the Arduino.

6. Operating Procedures

6.1 Normal Cycle

1. Set Hold Time selector switch (pin 11) to desired position before starting.
2. Power on the Arduino. Confirm serial monitor shows "DC Power Supply Controller" and the correct hold time.
3. Press the Start button. Relays 1–4 will energise in sequence (250 ms apart).
4. PWM ramps to 100% over 5 seconds.
5. Output holds at 100% for the selected duration. Press Start during this phase to reset the hold timer.
6. PWM ramps down to 0% over 5 seconds.
7. Relays de-energise in reverse order (4 → 1). A 5-second button lockout begins.

6.2 Test Mode

- Flip Test Mode switch (pin 8) to LOW while the system is idle.
- The cycle starts automatically — relay 2 is skipped throughout.
- The hold phase runs indefinitely until the switch is flipped back.
- Flipping the switch OFF triggers a graceful ramp-down from the current output level.

6.3 Emergency Shutdown

- Toggle the Shutdown switch (pin 3) LOW at any time.
- PWM is cut to 0 and all relays open immediately (no staged sequence).
- The system locks out for 15 seconds — the serial monitor counts down.
- After the lockout expires, hold the Shutdown switch LOW for 5 seconds to reactivate.
- Releasing early resets the reactivation timer — a full 5-second hold is required.

6.4 Reversing a Ramp-Down

If the Start button is pressed while the system is ramping down, the ramp direction reverses. The PWM level climbs back to 100% from its current position (it does not restart from 0%), and the hold timer resets to the full duration once 100% is reached.

7. Timing Reference

Parameter	Value	Notes
Relay stagger delay	250 ms	Between each relay turning on or off
Ramp-up time	5 seconds	0% → 100% PWM
Hold time (normal)	2 minutes	Pin 11 HIGH (open)
Hold time (short)	30 seconds	Pin 11 LOW (closed)
Ramp-down time	5 seconds	100% → 0% PWM
Button lockout after cycle	5 seconds	Prevents immediate restart
Shutdown lockout	15 seconds	After emergency shutdown triggered
Reactivation hold time	5 seconds	Must hold shutdown switch continuously
PWM update interval	50 ms	Ramp smoothing resolution
Switch debounce window	20 ms	Applied to both start and shutdown switches
Serial debug interval	500 ms	Status printed during ramp and hold phases

8. State Machine Reference

The controller uses a six-state machine. All state transitions are described below.

State	Name	Description
STATE_OFF	Idle	System idle. PWM = 0. All relays off. Waiting for start button.
STATE_RELAY_ON	Relay Staging On	Energises relays 1–4 in sequence, 250 ms apart.
STATE_RAMP_UP	Ramp Up	PWM increases linearly from 0 to 100% over 5 seconds.
STATE_HOLD	Hold	PWM holds at 100%. Timer counts down. In test mode, holds indefinitely.
STATE_RAMP_DOWN	Ramp Down	PWM decreases linearly from 100% to 0% over 5 seconds.
STATE_RELAY_OFF	Relay Staging Off	De-energises relays 4–1 in sequence, 250 ms apart. Returns to STATE_OFF.

Normal flow: STATE_OFF → STATE_RELAY_ON → STATE_RAMP_UP → STATE_HOLD → STATE_RAMP_DOWN → STATE_RELAY_OFF → STATE_OFF

9. Serial Monitor Messages

Connect at 9600 baud. Messages are printed for all state transitions and safety events.

Message	Meaning
DC Power Supply Controller	Startup confirmation
Hold time: Xs	Active hold duration read from pin 11 at startup
Starting relay ON sequence...	Relay staging sequence beginning
Relay N ON / OFF	Individual relay state change
Relay 2 SKIPPED (test mode)	Relay 2 bypassed due to test mode switch
Starting RAMP UP sequence (5 seconds)	Ramp-up phase started
Ramping UP: XX%	Live ramp progress (every 500 ms)
At maximum output (100%) - HOLDING...	Hold phase started, duration shown
HOLDING at 100% - Xs remaining	Hold countdown (every 500 ms)
Hold timer RESET - Xs remaining	Hold timer restarted by button press
Starting RAMP DOWN sequence...	Ramp-down phase started
All relays OFF - Cycle complete	Cycle finished, entering STATE_OFF
Button locked for X more seconds	Lockout active, time remaining shown
!!! SHUTDOWN TRIGGERED !!!	Emergency shutdown activated
Shutdown lockout: Xs remaining	15-second lockout countdown (every 1 s)
Hold to reactivate: Xs remaining	Reactivation hold countdown (every 1 s)
System REACTIVATED	System ready after successful reactivation hold

10. Key Constants — Quick Edit Reference

All timing and threshold values are defined as named constants near the top of the sketch. To adjust behaviour, change the value and re-upload — no other code changes are needed.

Constant	Default	Purpose
rampUpTime	5000 ms	Duration of the 0→100% PWM ramp
holdTimeLong	120000 ms	Hold duration when pin 11 is HIGH
holdTimeShort	30000 ms	Hold duration when pin 11 is LOW
rampDownTime	5000 ms	Duration of the 100→0% PWM ramp
relayStaggerDelay	250 ms	Delay between each relay switching
buttonLockoutTime	5000 ms	Post-cycle button lockout duration
shutdownLockoutTime	15000 ms	Lockout after emergency shutdown
reactivateHoldTime	5000 ms	Hold duration required to reactivate
debounceDelay	20 ms	Start button debounce window
shutdownDebounceDelay	20 ms	Shutdown switch debounce window
updateInterval	50 ms	PWM update rate during ramps
maxOutputLevel	255	Maximum PWM duty cycle (0–255)

11. Troubleshooting

Symptom	Likely Cause / Fix
Start button does nothing	Check 5-second button lockout (serial monitor will say so). Check pin 2 wiring. Confirm shutdown is not active.
Relay 2 never turns on	Test mode switch (pin 8) may be LOW. Flip it to HIGH (open).
Hold time is wrong	Check pin 11 switch position. The hold time is latched at cycle start — changing it mid-cycle has no effect.
System stuck after shutdown	Wait for the 15-second lockout to expire (serial monitor counts down), then hold the shutdown switch LOW for a full 5 seconds.
PWM output not reaching 100%	Confirm pin 9 is connected to a PWM-capable output device. Verify maxOutputLevel is 255.
Relays click erratically at startup	Shutdown switch may be reading LOW at power-on. Ensure it is in the open position before powering the board.
Serial monitor shows no output	Confirm baud rate is set to 9600. Check USB connection.
Ramp reversal doesn't feel smooth	Normal — the reversal back-dates the ramp start time to resume from the current level rather than restarting from 0.

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